MSQF 542 FINANCIAL ENGINEERING AND DERIVATIVES

UNIT IV – OPTIONS

INTRODUCTION TO OPTIONS CONTRACTS

Meaning of an Option

An option is a financial derivative contract that gives the buyer the right, but not the obligation, to buy or sell an underlying asset at a predetermined price on or before a specified date. In simple words, an option gives an investor the opportunity to buy or sell an asset at an agreed price in the future without forcing the investor to complete the transaction.

The underlying asset may be shares of a company, stock market indices such as NIFTY 50 and SENSEX, currencies such as USD/INR, commodities such as gold and crude oil, or financial instruments such as bonds and futures contracts.

Example: Suppose Reliance Industries shares trade at ₹1,300. An investor expects the price to rise and purchases a one-month call option with a strike price of ₹1,350 and premium of ₹30. If the share price reaches ₹1,450, the investor may exercise the right to buy at ₹1,350. If the price falls to ₹1,200, the investor may allow the option to expire. The maximum loss for the buyer is the ₹30 premium per share.

Key idea: An option buyer has a right, not an obligation. The option seller receives the premium and accepts the corresponding contractual obligation.

1.2 Important Terminologies in Options

Term	Explanation / Illustration
Underlying Asset	The financial instrument on which the option contract is based. For a NIFTY option, the NIFTY 50 index is the underlying.
Strike Price (Exercise Price)	The predetermined price at which the option buyer has the right to buy or sell the underlying asset.
Option Premium	The amount paid by the option buyer to the seller. If the premium is ₹100 and contract quantity is 50, the premium paid is ₹100 × 50 = ₹5,000.
Expiry Date	The final date on which the option contract remains valid. At expiry, the contract is settled according to its terms or expires worthless.
Option Buyer (Holder)	The person who purchases the option and pays the premium. The buyer has a right, no obligation to exercise, and generally risks the premium paid.

Option Seller (Writer)	The person who sells the option and receives the premium. The seller may have to meet the contract terms if assigned and must maintain applicable margin.
Lot Size	The number of units covered by one derivative contract. If one lot contains 50 units and premium is ₹40, the total premium is ₹2,000.

1.3 Call Option and Put Option

A call option gives the buyer the right to purchase the underlying asset at the strike price. It is generally purchased when the investor expects the market price to increase.

A put option gives the buyer the right to sell the underlying asset at the strike price. It is generally purchased when the investor expects the market price to decrease.

Call option numerical illustration

Current share price = ₹500; strike price = ₹520; call premium = ₹15; expiry = one month.

Share price at expiry	Exercise decision	Profit / Loss per share
₹560	Exercise	₹40 intrinsic value − ₹15 premium = +₹25
₹535	Exercise	₹15 intrinsic value − ₹15 premium = ₹0
₹510	Do not exercise	−₹15

Call break-even = Strike price + Premium = ₹520 + ₹15 = ₹535

If the share price is ₹560, intrinsic value is ₹560 − ₹520 = ₹40. Net profit is ₹40 − ₹15 = ₹25 per share. The investor begins earning a net profit only above ₹535 at expiry.

Put option numerical illustration

Current share price = ₹500; strike price = ₹480; put premium = ₹12.

Share price at expiry	Exercise decision	Profit / Loss per share
₹430	Exercise	₹50 intrinsic value − ₹12 premium = +₹38
₹468	Exercise	₹12 intrinsic value − ₹12 premium = ₹0
₹490	Do not exercise	−₹12

Put break-even = Strike price − Premium = ₹480 − ₹12 = ₹468

At a share price of ₹430, put intrinsic value is ₹480 − ₹430 = ₹50. Net profit is ₹50 − ₹12 = ₹38 per share.

2. TYPES OF OPTIONS

2.1 European Options

A European option can be exercised only on its expiry date. Example: An investor purchases a European call with strike ₹1,000 expiring on 30 December. Even if the market price reaches ₹1,200 on 15 December, the investor cannot exercise it on that date; exercise is permitted only on expiry.

2.2 American Options

An American option can be exercised at any time up to and including its expiry date. Example: An American call with strike ₹500 may be exercised when the share price reaches ₹600 ten days before expiry, or the investor may wait.

European Option	American Option
Exercise only on expiry	Exercise any time up to expiry
Restricted exercise timing	Flexible exercise timing
Often simpler to value in standard models	Early-exercise possibility must be considered
Common in many index and stock derivative markets	Common in certain exchange-traded and equity option markets

The terms European and American describe exercise style, not necessarily the geographical market in which the options are traded.

2.4 Exchange-Traded Options

These options are traded through recognised exchanges. They generally have standardised contract specifications, expiry dates and strike prices, clearing-house settlement, margin requirements for sellers and greater transparency. NIFTY and BANK NIFTY options traded on NSE are examples.

2.5 Over-the-Counter (OTC) Options

OTC options are privately negotiated contracts between two parties, often through financial institutions. Contract quantity, strike price, expiry date, settlement conditions and underlying asset can be customised. Example: An Indian company expecting USD 1 million after six months may enter a customised currency option agreement with a bank to protect against adverse exchange-rate movements.

3. OPTION PAYOFF AND PROFIT CALCULATION

3.1 Difference Between Payoff and Profit

Payoff is the value received from exercising an option at expiry. Profit or loss is the payoff after deducting the premium paid by a buyer or adding the premium received by a seller.

3.2 Call Option Buyer Payoff

$$\text{Call Payoff} = \max(S_t - K, 0)$$

$$\text{Call Buyer Profit} = \max(S_t - K, 0) - C$$

S_t is the market price at expiry, K is the strike price and C is the call premium paid.

Example: Strike price ₹100 and premium ₹10.

Expiry share price	Payoff	Net profit / loss
₹80	₹0	−₹10
₹90	₹0	−₹10
₹100	₹0	−₹10
₹105	₹5	−₹5
₹110	₹10	₹0
₹120	₹20	+₹10
₹130	₹30	+₹20

Below the strike price, the option is not exercised and the buyer loses the premium. The break-even is ₹110. Above ₹110, the buyer earns a net profit. Potential profit is theoretically unlimited because the underlying price can keep rising.

3.3 Call Option Seller Payoff

$$\text{Call Seller Profit} = C - \max(S_t - K, 0)$$

With strike ₹100 and premium received ₹10:

Expiry price	Seller profit / loss
₹80	+₹10
₹90	+₹10
₹100	+₹10
₹105	+₹5
₹110	₹0
₹120	−₹10
₹130	−₹20

The maximum profit is the premium received. An uncovered call writer faces theoretically unlimited loss.

3.4 Put Option Buyer Payoff

$$\text{Put Payoff} = \max(K - S_t, 0)$$

$$\text{Put Buyer Profit} = \max(K - S_t, 0) - P$$

With strike ₹100 and premium ₹8:

Expiry price	Payoff	Net profit / loss
₹120	₹0	−₹8
₹110	₹0	−₹8
₹100	₹0	−₹8
₹95	₹5	−₹3
₹92	₹8	₹0
₹80	₹20	+₹12
₹70	₹30	+₹22

The maximum loss is the premium paid. For an ordinary share whose price cannot fall below zero, maximum profit is limited to the strike price less the premium.

3.5 Put Option Seller Payoff

$$\text{Put Seller Profit} = P - \max(K - S_t, 0)$$

With strike ₹100 and premium received ₹8, if the share price falls to ₹70, loss = (₹100 − ₹70) − ₹8 = ₹22 per share. If the share price remains above ₹100, the seller retains the ₹8 premium.

3.6 Summary of Four Basic Positions

Position	Market expectation	Maximum profit	Maximum loss
Buy Call	Bullish	Unlimited	Premium paid
Sell Call	Neutral / bearish	Premium received	Unlimited if uncovered
Buy Put	Bearish	Strike less premium (subject to underlying floor)	Premium paid
Sell Put	Neutral / bullish	Premium received	Substantial; depends on underlying price

4. FACTORS AFFECTING OPTION PRICES

An option premium depends on several factors, not only the current market price of the underlying asset.

Underlying asset price: An increase in the share price generally increases a call's value and decreases a put's value, other things being equal.

Strike price: The relationship between market price and strike price determines whether an option is in-the-money, at-the-money or out-of-the-money.

Time to expiry: More time generally increases an option's time value because there is more time for a favourable movement.

Volatility: Higher volatility generally increases the values of both calls and puts because the possibility of a favourable movement becomes greater.

Risk-free interest rate: A higher risk-free rate generally increases call values and decreases put values, assuming other factors remain constant and no dividends.

Dividend: Expected dividends generally reduce call values and increase put values, all else equal, because the underlying stock price is expected to adjust downward when a dividend is paid.

4.1 Option Greeks – Basic Introduction

Option Greeks measure how sensitive an option's value is to changes in market factors.

Greek	Meaning	Interpretation
Delta	Sensitivity to underlying price	Approximate option-value change when underlying price changes by ₹1.
Gamma	Sensitivity of delta	How much delta changes when underlying price changes.
Theta	Sensitivity to time	Effect of passage of time on option premium.
Vega	Sensitivity to volatility	Effect of a change in implied volatility.
Rho	Sensitivity to interest rate	Effect of a change in interest rates.

Delta example: If a call has delta 0.60 and the underlying rises by ₹10, the option premium may rise approximately ₹6, assuming other factors remain unchanged and the movement is sufficiently small.

Theta example: If theta is −2, the option may lose approximately ₹2 per day due to time passing, all else equal. This is called time decay.

5. OPTION PRICING MODELS

5.1 Meaning of Option Pricing

Option pricing is the process of estimating the fair theoretical value of an option. Traders and analysts use mathematical models to assess whether an option is fairly priced, overpriced or underpriced. The two models in this unit are the Binomial Option Pricing Model and the Black-Scholes Option Pricing Model.

5.2 Binomial Option Pricing Model

The Binomial Option Pricing Model estimates an option's theoretical value by assuming that the underlying asset can move in one of two directions during each time interval: upward or downward. By calculating possible future option payoffs and discounting their risk-neutral expected value, the model estimates the current option price.

Basic assumptions

- The underlying asset price can move upward or downward during each time interval.
- The model uses a specified up factor and down factor.
- A risk-free interest rate is available.
- There are no arbitrage opportunities.
- The option's future payoff depends on the underlying asset price.
- Market and contract assumptions remain consistent during the calculation.

Structure and formulas

Suppose the current share price is ₹100. After one year it may rise to ₹120 or fall to ₹90. This is represented by a one-period binomial tree.

$$S_u = S_0 \times u \quad \text{and} \quad S^d = S_0 \times d$$

S_0 is the current share price, S_u and S^d are the up and down prices, and u and d are the up and down factors.

Numerical problem – one-period Binomial Model

A share is currently trading at ₹100. After one year, its price may increase by 20% or decrease by 10%. A call option has strike price ₹100. The annual risk-free interest rate is 5%. Calculate the theoretical call price.

Particular	Value
Current share price (S_0)	₹100
Up factor (u)	1.20
Down factor (d)	0.90
Strike price (K)	₹100
Risk-free rate (r)	5%
Time	1 year

Step 1 – Future share prices:

$$S_u = 100 \times 1.20 = ₹120 \quad S^d = 100 \times 0.90 = ₹90$$

Step 2 – Call payoffs:

$$C_u = \max(120 - 100, 0) = ₹20 \quad C^d = \max(90 - 100, 0) = ₹0$$

Step 3 – Risk-neutral probability:

$$q = [(1 + r) - d] / (u - d)$$

$$q = (1.05 - 0.90) / (1.20 - 0.90) = 0.15 / 0.30 = 0.50$$

The downward probability is $1 - q = 0.50$.

Step 4 – Present option value:

$$C_0 = [qC_u + (1 - q)C^d] / (1 + r)$$

$$C_0 = [(0.50 \times 20) + (0.50 \times 0)] / 1.05 = 10 / 1.05 = ₹9.52$$

Final answer: The theoretical fair value of the call option is approximately ₹9.52. A market premium below or above this model value may warrant investigation, but the comparison depends on suitable inputs and assumptions.

Advantages

- Easy to understand and implement.
- Can accommodate American options and early exercise.
- Can handle different exercise dates and more complex features.
- Can be expanded into multiple time periods.
- Useful for teaching and practical financial modelling.

Limitations

- Calculations become lengthy when many periods are used.
- Results depend on assumptions about upward and downward movements.
- Complex contracts may require substantial computation.
- Incorrect inputs can produce misleading theoretical values.

5.3 Multi-Period Binomial Model

The one-period model considers only one upward or downward movement. A multi-period model divides the option's life into several intervals. With $S_0 = ₹100$, $u = 1.20$ and $d = 0.90$, after two periods:

Path	Calculation	Terminal price
Up–Up	100 × 1.20 × 1.20	₹144
Up–Down or Down–Up	100 × 1.20 × 0.90	₹108
Down–Down	100 × 0.90 × 0.90	₹81

The same intermediate price ₹108 is reached through an up movement followed by a down movement or the reverse. This is a recombining binomial tree. The model first calculates terminal option payoffs and then works backward to find the present value. This is called backward induction.

$$V = [qV_u + (1 - q)V^d] / (1 + r\Delta t)$$

V is the option value at a node; V_u and V^d are values at the next up and down nodes; q is the risk-neutral probability; and $r\Delta t$ represents the risk-free return for the interval. For an American option, each node takes the greater of immediate exercise value and continuation value:

$$V = \max(\text{Exercise Value, Continuation Value})$$

This feature makes the binomial model particularly useful for valuing American-style options.

6. BLACK-SCHOLES OPTION PRICING MODEL

6.1 Meaning

The Black-Scholes Option Pricing Model, developed by Fischer Black and Myron Scholes and extended by Robert Merton, is widely used to calculate the theoretical value of European call and put options. It uses the current underlying price, strike price, time to expiry, risk-free rate and volatility.

6.2 Assumptions

- The option is European-style.
- The underlying asset price follows a continuous stochastic process with a lognormal distribution.
- Volatility remains constant throughout the option's life.
- The risk-free interest rate remains constant.
- Markets are frictionless, with no transaction costs or taxes.
- Trading takes place continuously.
- No arbitrage opportunities exist.
- The basic model assumes the underlying asset does not pay dividends. A dividend-adjusted version is available.

6.3 BLACK-SCHOLES FORMULAS

$$\text{Call: } C = S_0N(d_1) - Ke^{\wedge}(-rT)N(d_2)$$

$$\text{Put: } P = Ke^{\wedge}(-rT)N(-d_2) - S_0N(-d_1)$$

$$d_1 = [\ln(S_0/K) + (r + \sigma^2/2)T] / (\sigma\sqrt{T})$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Symbol	Meaning
C / P	Theoretical call / put price
S_0	Current underlying asset price
K	Strike price

r	Continuously compounded risk-free rate
T	Time to expiry in years
σ	Annualised volatility
e	Mathematical constant
N(d)	Cumulative standard normal distribution function

6.4 Numerical problem – Black-Scholes call option

The current share price is ₹100. A European call has strike ₹100 and expires after one year. The annual risk-free rate is 5% and annual volatility is 20%. Calculate the theoretical call value.

Particular	Value
S_0	₹100
K	₹100
r	0.05
σ	0.20
T	1 year

Step 1 – Calculate d_1:

$$d_1 = [\ln(100/100) + (0.05 + 0.20^2/2) \times 1] / (0.20 \times \sqrt{1})$$

$$d_1 = (0 + 0.05 + 0.02) / 0.20 = 0.35$$

Step 2 – Calculate d_2:

$$d_2 = 0.35 - (0.20 \times 1) = 0.15$$

Step 3 – Obtain cumulative standard normal values:

$$N(0.35) = 0.6368 \quad N(0.15) = 0.5596$$

Step 4 – Calculate call price:

$$C = 100(0.6368) - 100e^{\wedge}(-0.05)(0.5596)$$

$$e^{\wedge}(-0.05) \approx 0.95123$$

$$C = 63.68 - (100 \times 0.95123 \times 0.5596)$$

$$C \approx 63.68 - 53.23 = ₹10.45$$

Final answer: The theoretical European call option price is approximately ₹10.45.

Interpretation: Under the model assumptions, the estimated fair premium is ₹10.45. A difference between market premium and model value may arise from implied volatility, demand and supply, transaction costs, liquidity or deviations from model assumptions.

6.5 Advantages

- Provides a systematic approach to option valuation.
- Widely accepted in financial markets.
- Useful for estimating theoretical option premiums and implied volatility.
- Provides the foundation for calculating option Greeks.
- Useful in portfolio risk management and financial engineering.

6.6 Limitations

- Assumes constant volatility and a constant risk-free interest rate.
- The traditional model applies to European options.
- Does not fully reflect sudden market shocks.
- Actual markets may experience jumps and changing volatility.
- Transaction costs and liquidity constraints are excluded from the basic model.

6.7 Binomial Model versus Black-Scholes Model

Basis	Binomial Model	Black-Scholes Model
Approach	Discrete-time model	Continuous-time model
Price movement	Upward or downward movements	Continuous price movement
Structure	Tree-based	Mathematical formula
Exercise style	Can value European and American options	Traditional model is for European options
Flexibility	More flexible for early exercise	Less flexible in basic form
Calculation	Backward induction	Formula-based
Complexity	Depends on number of periods	Direct once inputs are available
Applications	Option valuation and financial engineering	Option valuation and risk management

As the number of binomial periods increases, the binomial model can approach the Black-Scholes value under suitable assumptions. They are different mathematical approaches to the same valuation problem.

7. DIFFERENCES BETWEEN FUTURES AND OPTIONS CONTRACTS

Futures versus Options

Basis	Futures Contract	Options Contract
Meaning	Agreement to buy or sell	Right to buy or sell for buyer
Obligation	Both parties have obligations	Buyer has a right; seller has an obligation
Premium	No option premium	Buyer pays premium
Margin	Both sides generally maintain margins	Option seller generally maintains margin
Risk to buyer	Can be substantial	Limited to premium paid by option buyer
Profit potential	Depends on price movement	Depends on option type and position
Time value	No option time value	Option has time value
Exercise / settlement	Settled according to contract terms	Depends on exercise style and settlement terms
Market exposure	Directional exposure	Directional, hedging and volatility exposure
Strategies	Long and short futures	Multiple combinations of calls and puts

A share trades at ₹500 and an investor expects its price to increase.

Choice A – Buy Futures: Buy futures at ₹500. If price rises to ₹550, profit = ₹550 − ₹500 = ₹50. If it falls to ₹450, loss = ₹450 − ₹500 = −₹50.

Choice B – Buy a Call: Buy a call with strike ₹500 and premium ₹20. If price rises to ₹550, profit = (₹550 − ₹500) − ₹20 = ₹30. If price falls to ₹450, loss is limited to ₹20 premium.

Futures provide direct exposure to price movements. Options provide more flexible payoff structures and, for a buyer, a defined premium risk.

8. OPTIONS TRADING STRATEGIES

Options trading strategies combine one or more calls and puts to achieve an investment objective: benefit from a rising or falling market, earn premium income, protect an existing portfolio, benefit from high or low volatility, limit losses or create a specific risk-reward profile.

- Covered Call
- Protective Put
- Spreads: Bull Spread, Bear Spread, Butterfly Spread and Calendar Spread
- Straddle
- Strips and Straps
- Strangle

9. COVERED CALL STRATEGY

9.1 Meaning

A covered call is a strategy in which an investor owns the underlying shares and sells a call option against those shares. The investor receives premium and accepts the obligation to sell the shares at the strike price if assigned. It is covered because the investor already owns the shares needed to meet the delivery obligation.

9.2 Construction

- Purchase or already own the underlying shares.
- Sell one call option against the shares held.
- Receive the option premium.

9.3 Numerical illustration

An investor purchases 100 shares at ₹500 each and sells a call with strike ₹550, receiving ₹20 premium per share.

Particular	Value
Share purchase price	₹500
Strike price	₹550
Call premium received	₹20
Quantity	100 shares
Initial share investment	₹50,000
Premium received	₹2,000

Share price at expiry	Share P/L	Premium received	Total P/L
₹450	−₹5,000	₹2,000	−₹3,000
₹480	−₹2,000	₹2,000	₹0
₹500	₹0	₹2,000	+₹2,000
₹530	+₹3,000	₹2,000	+₹5,000

₹550	+₹5,000	₹2,000	+₹7,000
₹600	+₹5,000*	₹2,000	+₹7,000

*At ₹600, the shares are assumed to be called away at the strike price ₹550; therefore, the share gain is capped at ₹50 per share.

$$\text{Covered call profit} = (S_t - S_0) - \max(S_t - K, 0) + C$$

S_t is share price at expiry, S_0 is purchase price, K is call strike and C is premium received.

Advantages

- Generates additional income from premium.
- Provides a small cushion against a share-price decline.
- Suitable for stable or moderately rising markets.
- Can improve income from an existing share portfolio.

Limitations

- Upside profit is limited.
- The investor remains exposed to substantial downside risk.
- Shares may be sold at the strike even if market price rises significantly.
- Premium may not fully protect against a large decline.

Practical application: A long-term investor expecting limited price movement over the next month may consider a covered call, while accepting the possibility of the shares being called away.

10. PROTECTIVE PUT STRATEGY

10.1 Meaning

A protective put is a strategy in which an investor owns shares and purchases a put option on the same underlying asset. The put acts like insurance by giving the investor the right to sell at the strike price, subject to contract terms, even if the market price falls.

10.2 Construction

- Buy or hold the underlying shares.
- Purchase a put option on the same shares.

10.3 Numerical illustration

An investor purchases 100 shares at ₹500 and buys a put with strike ₹480 for a premium of ₹15.

Particular	Value
Share investment	₹50,000
Put premium	₹1,500
Total initial cost	₹51,500

Share price at expiry	Share P/L	Put payoff	Net P/L
₹400	−₹10,000	₹8,000	−₹3,500
₹450	−₹5,000	₹3,000	−₹500
₹480	−₹2,000	₹0	−₹3,500
₹500	₹0	₹0	−₹1,500
₹550	+₹5,000	₹0	+₹3,500
₹600	+₹10,000	₹0	+₹8,500

At ₹400, put payoff is ₹480 − ₹400 = ₹80 per share, or ₹8,000 for 100 shares. The share loss is ₹10,000, so the total loss after the put payoff and premium is ₹3,500.

Protective put profit = $(S_t - S_0) + \max(K - S_t, 0) - P$

Maximum loss per share = $S_0 - K + P = 500 - 480 + 15 = ₹35$

Advantages

- Protects against a significant decline.
- Allows the investor to continue holding shares.
- Limits downside risk to a known amount, subject to contract terms.
- Maintains upside potential, less the put premium.

Limitations

- The investor must pay the put premium.
- Protection is limited to the strike and expiry.
- Repeated put purchases can reduce long-term returns.

Practical application: An investor holding shares before quarterly results may purchase a protective put to reduce the impact of a sudden adverse price movement.

11. SPREAD STRATEGIES

11.1 Meaning of Spread

A spread is an option strategy in which an investor simultaneously buys and sells two or more options on the same underlying asset. The options generally differ in strike prices, expiry dates or both.

Spreads can reduce initial premium, limit maximum losses and profits, express a market outlook and manage volatility and time-decay exposure.

11.2 Types of Spreads

Type	Description
Vertical spread	Options have the same expiry but different strike prices.
Horizontal spread	Options have the same strike price but different expiry dates; also called a calendar or time spread.
Diagonal spread	Options have different strike prices and different expiry dates.

12. BULL SPREAD

12.1 Meaning

A bull spread is used when an investor expects the market price to increase moderately. It has limited maximum profit and limited maximum loss. The two common types are bull call spread and bull put spread.

12.2 Bull Call Spread

Construct a bull call spread by buying a call at a lower strike and selling another call at a higher strike. Both options have the same underlying asset and expiry.

Example: Current share price ₹500. Buy a ₹500 call for ₹30 and sell a ₹550 call for ₹10.

Particular	Buy Call	Sell Call
Strike price	₹500	₹550
Premium	₹30	₹10
Position	Long	Short

Net premium paid = 30 − 10 = ₹20

Maximum loss = Net debit = ₹20

Maximum profit = (550 − 500) − 20 = ₹30

Break-even = Lower strike + Net debit = 500 + 20 = ₹520

Share price at expiry	Long call P/L	Short call P/L	Net P/L
₹450	−₹30	+₹10	−₹20
₹500	−₹30	+₹10	−₹20
₹510	−₹20	+₹10	−₹10
₹520	−₹10	+₹10	₹0
₹530	₹0	+₹10	+₹10
₹550	+₹20	+₹10	+₹30
₹580	+₹50	−₹20	+₹30

Below ₹500 the loss is the net premium of ₹20. Between ₹500 and ₹550 profit increases. At ₹520 the spread breaks even. At ₹550 and above, maximum profit is ₹30.

Advantages

- Maximum loss is limited.
- Costs less than buying a call alone in many cases.
- Suitable for moderately bullish expectations.
- Break-even and maximum profit are known.

Limitations

- Profit is capped.
- The investor may miss large gains if the underlying rises sharply.

12.3 Bull Put Spread

A bull put spread is created by selling a put at a higher strike and buying another put at a lower strike, with the same expiry. It receives a net premium and benefits if the market remains stable or rises.

Particular	Sell Put	Buy Put
Strike price	₹500	₹450
Premium	₹25	₹10

Net premium received = 25 − 10 = ₹15

Maximum profit = ₹15

Maximum loss = (500 − 450) − 15 = ₹35

Break-even = 500 − 15 = ₹485

The maximum profit is earned if the share price remains at or above ₹500 at expiry.

13. BEAR SPREAD

13.1 Meaning

A bear spread is used when an investor expects the underlying asset price to decline moderately. It aims to profit from a fall while limiting maximum profit and loss. The two common types are bear put spread and bear call spread.

Bear Put Spread

Construct a bear put spread by buying a put at a higher strike and selling another put at a lower strike, with the same expiry.

Example: Current share price ₹500. Buy the ₹500 put for ₹30 and sell the ₹450 put for ₹10.

Particular	Buy Put	Sell Put
Strike price	₹500	₹450
Premium	₹30	₹10

Net premium paid = 30 − 10 = ₹20

Maximum loss = ₹20

Maximum profit = (500 − 450) − 20 = ₹30

Break-even = 500 − 20 = ₹480

Share price at expiry	Buy Put P/L	Sell Put P/L	Net P/L
₹400	+₹70	−₹40	+₹30
₹430	+₹40	−₹10	+₹30
₹450	+₹20	+₹10	+₹30
₹480	−₹10	+₹10	₹0
₹500	−₹30	+₹10	−₹20
₹550	−₹30	+₹10	−₹20

Below ₹450, maximum profit is ₹30. Between ₹450 and ₹500, profit declines as the share price rises. At ₹480 the spread breaks even. At ₹500 or above, maximum loss is ₹20.

Advantages

- Suitable for moderately bearish expectations.
- Risk is limited to the net premium paid.
- Maximum profit is known in advance.
- May cost less than purchasing an unhedged put.

Bear Call Spread

A bear call spread is created by selling a call at a lower strike and buying another call at a higher strike, with the same expiry.

Particular	Sell Call	Buy Call
Strike price	₹500	₹550
Premium	₹30	₹10

Net premium received = 30 − 10 = ₹20

Maximum profit = ₹20

Maximum loss = (550 − 500) − 20 = ₹30

Break-even = 500 + 20 = ₹520

The strategy earns maximum profit if the underlying asset remains at or below ₹500 at expiry.

BUTTERFLY SPREAD

14.1 Meaning

A butterfly spread is designed to generate profit when the underlying asset price remains near a target price at expiry. It is generally suitable when an investor expects low volatility or limited price movement. A standard long call butterfly uses three strike prices: lower, middle and higher.

Construction: Buy one lower-strike call, sell two middle-strike calls and buy one higher-strike call. All options have the same underlying and expiry.

14.2 Numerical illustration

Current share price ₹500. Buy one ₹450 call for ₹70; sell two ₹500 calls for ₹40 each; buy one ₹550 call for ₹20.

Position	Strike price	Premium
Buy 1 Call	₹450	₹70
Sell 2 Calls	₹500	₹40 each
Buy 1 Call	₹550	₹20

Net premium paid = 70 − (2 × 40) + 20 = ₹10

Maximum loss = ₹10

Maximum profit = Wing width − net debit = 50 − 10 = ₹40

Lower break-even = 450 + 10 = ₹460

Upper break-even = 550 − 10 = ₹540

Share price at expiry	Net payoff	Net profit / loss
₹400	₹0	−₹10
₹450	₹0	−₹10
₹460	₹10	₹0
₹475	₹25	+₹15
₹500	₹50	+₹40
₹525	₹25	+₹15
₹540	₹10	₹0
₹550	₹0	−₹10
₹600	₹0	−₹10

Maximum profit occurs at the middle strike price of ₹500. Profit declines as the share price moves away from ₹500.

Advantages

- Maximum loss is limited.
- Useful in a low-volatility market.
- May require a relatively small initial investment.
- Maximum profit and break-even points are known.

Limitations

- Profit is concentrated around the middle strike.
- A large market movement can result in maximum loss.
- Strike prices must be selected carefully.

15. CALENDAR SPREAD

15.1 Meaning

A calendar spread involves buying and selling options with the same strike price but different expiry dates. A common construction is to sell a near-term option and buy a longer-term option. It is also called a time spread and attempts to benefit from differences in time decay and volatility.

15.2 Numerical illustration

Current share price ₹500. Sell a near-month ₹500 call for ₹15 and buy a far-month ₹500 call for ₹25.

Position	Strike	Expiry	Premium
Sell Call	₹500	Near month	₹15
Buy Call	₹500	Far month	₹25

Net premium paid = ₹25 − ₹15 = ₹10

Suppose at the near-month expiry the short call expires worthless and the remaining long call is worth ₹18. The remaining option value is ₹18; compared with the initial net cost of ₹10, the indicative position value is ₹8 higher. This is an unrealised mark-to-market gain at the near-month expiry, not necessarily the final realised profit, because the long option still has time remaining.

Advantages

- Can benefit from faster time decay of the near-term option.
- May be useful when the investor expects the market to remain near a selected price in the short term.
- The longer-term option provides continued exposure.

Limitations

- Sensitive to changes in implied volatility.
- May lose if the underlying moves sharply.

- Requires monitoring when the near-term option expires.
- Profit cannot be determined from the initial premium difference alone.

Practical application: An investor expects a stock to remain near ₹500 until the current monthly expiry but expects greater movement over the following month. A calendar spread may express this view.

STRADDLE STRATEGY

16.1 Meaning and construction

A straddle involves buying a call and a put on the same underlying asset with the same strike price and expiry date. A long straddle is used when an investor expects a large price movement but is uncertain about its direction.

Construction: Buy one call and one put, using the same strike and expiry.

16.2 Numerical illustration

Share price ₹500. Buy a ₹500 call for ₹20 and a ₹500 put for ₹15.

Option	Strike price	Premium
Call	₹500	₹20
Put	₹500	₹15

Total premium = 20 + 15 = ₹35 (maximum loss)

Upper break-even = 500 + 35 = ₹535

Lower break-even = 500 − 35 = ₹465

Share price at expiry	Call payoff	Put payoff	Net profit / loss
₹400	₹0	₹100	+₹65
₹450	₹0	₹50	+₹15
₹465	₹0	₹35	₹0
₹480	₹0	₹20	−₹15
₹500	₹0	₹0	−₹35
₹520	₹20	₹0	−₹15
₹535	₹35	₹0	₹0
₹550	₹50	₹0	+₹15
₹600	₹100	₹0	+₹65

If the price remains at ₹500, both options may expire worthless and the investor loses ₹35. If it rises to ₹600 or falls to ₹400, the payoff is ₹100 and net profit is ₹65.

Advantages

- Profits from a large upward or downward movement.
- Does not require predicting the direction.
- Maximum loss is limited to total premium.
- May be useful before major events if a large move is expected.

Limitations

- A sufficiently large movement is needed to cover both premiums.
- The position suffers from time decay.
- A stable market can cause loss of the full premium.
- High implied volatility can make the position expensive.

17. STRIPS AND STRAPS

Strips and straps are variations of the long straddle. Both involve buying calls and puts with the same strike price and expiry. The difference is the number of calls and puts purchased.

17.1 Strip Strategy

A strip consists of buying one call and two puts with the same strike and expiry. It is useful when a large movement is expected but a downward movement is considered more likely.

Example: Share price ₹500; call premium ₹20; put premium ₹15. Buy one call and two puts.

Total premium = 20 + (2 × 15) = ₹50

Lower break-even = 500 − (50 / 2) = ₹475

Upper break-even = 500 + 50 = ₹550

Share price at expiry	Call payoff	Two put payoffs	Net P/L
₹400	₹0	₹200	+₹150
₹450	₹0	₹100	+₹50
₹475	₹0	₹50	₹0
₹500	₹0	₹0	−₹50
₹525	₹25	₹0	−₹25
₹550	₹50	₹0	₹0
₹600	₹100	₹0	+₹50

The strip earns greater profits from a downward movement because the investor holds two puts.

17.2 Strap Strategy

A strap consists of buying two calls and one put with the same strike and expiry. It is useful when a large movement is expected but an upward movement is considered more likely.

Example: Share price ₹500; call premium ₹20; put premium ₹15. Buy two calls and one put.

Total premium = (2 × 20) + 15 = ₹55

Lower break-even = 500 − 55 = ₹445

Upper break-even = 500 + (55 / 2) = ₹527.50

Share price at expiry	Two call payoffs	Put payoff	Net P/L
₹400	₹0	₹100	+₹45

₹445	₹0	₹55	₹0
₹450	₹0	₹50	−₹5
₹500	₹0	₹0	−₹55
₹525	₹50	₹0	−₹5
₹527.50	₹55	₹0	₹0
₹550	₹100	₹0	+₹45
₹600	₹200	₹0	+₹145

The strap provides a greater payoff when the market rises because the investor holds two calls.

17.3 Straddle, Strip and Strap – Comparison

Basis	Straddle	Strip	Strap
Call options	1	1	2
Put options	1	2	1
Market expectation	Large movement either direction	Large movement, bearish bias	Large movement, bullish bias
Downside exposure	One put	Two puts	One put
Upside exposure	One call	One call	Two calls
Premium	Depends on call and put premiums	Higher than equivalent straddle if call and put premiums are equal	Higher than equivalent straddle if call and put premiums are equal

18. STRANGLE STRATEGY

18.1 Meaning

A strangle involves buying an out-of-the-money call and an out-of-the-money put on the same underlying asset with the same expiry date. Unlike a straddle, the call and put have different strike prices. A long strangle is used when a substantial movement in either direction is expected and the investor wants a lower initial premium than a comparable straddle.

18.2 Long strangle – numerical illustration

Current share price ₹500. Buy a ₹450 put for ₹10 and a ₹550 call for ₹12.

Option	Strike price	Premium
Put	₹450	₹10
Call	₹550	₹12

Total premium = ₹22

Lower break-even = 450 − 22 = ₹428

Upper break-even = 550 + 22 = ₹572

Share price at expiry	Put payoff	Call payoff	Net P/L
₹400	₹50	₹0	+₹28
₹428	₹22	₹0	₹0
₹430	₹20	₹0	−₹2
₹450	₹0	₹0	−₹22
₹500	₹0	₹0	−₹22
₹550	₹0	₹0	−₹22
₹572	₹0	₹22	₹0
₹600	₹0	₹50	+₹28

Between ₹450 and ₹550, both options expire worthless and the maximum loss is ₹22. Below ₹428 or above ₹572, the long strangle starts making a net profit.

Straddle versus Strangle

Basis	Straddle	Strangle
Strike prices	Same	Different
Call	Usually at-the-money	Out-of-the-money
Put	Usually at-the-money	Out-of-the-money
Initial premium	Usually higher	Usually lower
Movement required	A smaller movement may be sufficient	A larger movement is generally required
Market outlook	High volatility	High volatility
Risk for long position	Limited to premium	Limited to premium

19. PUT-CALL PARITY THEOREM

19.1 Meaning

Put-Call Parity is a principle explaining the relationship between European call and put prices when they have the same underlying asset, strike price and expiry date. The combined value of a call and the present value of the strike price must equal the combined value of the underlying asset and a put. The relationship prevents arbitrage under the theorem's assumptions.

19.2 Formula

$$C + Ke^{(-rT)} = P + S_0$$

$$C - P = S_0 - Ke^{(-rT)}$$

C is the European call price; P is the European put price; S_0 is the current underlying price; K is strike price; r is the risk-free interest rate; T is time to expiry; and e is the mathematical constant.

19.3 Why does Put-Call Parity exist?

Consider two portfolios. Portfolio A contains a call option and an investment equal to the present value of the strike price. Portfolio B contains one share of the underlying asset and one put option. At expiry, both portfolios have the same value in every price scenario. Therefore, their current values must be equal under the no-arbitrage principle.

Illustration: Share price ₹100, strike ₹100, one-year expiry and risk-free rate 5%.

Situation 1 – Share price rises to ₹120

Portfolio A: Call payoff = ₹120 − ₹100 = ₹20; risk-free investment grows to ₹100; total = ₹120.
Portfolio B: Share value = ₹120; put payoff = ₹0; total = ₹120.

Situation 2 – Share price falls to ₹80

Portfolio A: Call payoff = ₹0; risk-free investment = ₹100; total = ₹100. Portfolio B: Share value = ₹80; put payoff = ₹100 − ₹80 = ₹20; total = ₹100.

Both portfolios produce identical values in both situations. Therefore, they must have the same current value.

19.4 Numerical problem – Put-Call Parity

Current share price ₹100; European call premium ₹10.45; strike ₹100; risk-free rate 5%; expiry one year. Calculate the theoretical put premium.

Particular	Value
Share price	₹100
Call premium	₹10.45
Strike price	₹100
Interest rate	5%
Time to expiry	1 year

$$P = C + Ke^{(-rT)} - S_0$$

$$P = 10.45 + 100e^{(-0.05)} - 100$$

$$P = 10.45 + 95.12 - 100 = ₹5.57$$

Final answer: The theoretical European put premium is approximately ₹5.57. A market difference may indicate a pricing inconsistency, but transaction costs, liquidity, dividends and exercise restrictions must be considered before identifying an actual arbitrage opportunity.

Importance

- Helps identify mispriced options.
- Supports arbitrage analysis.
- Helps calculate an unknown call or put premium.
- Explains the relationship between calls and puts.
- Helps construct synthetic financial positions.
- Useful in financial engineering and risk management.

PRACTICAL APPLICATIONS OF OPTIONS IN FINANCIAL MARKETS

Hedging Against Market Risk

Investors use options to protect their investments against adverse price movements. Example: An investor holds shares worth ₹5 lakh and expects a possible market correction. Purchasing put options can provide downside protection.

Generating Additional Income

Investors who hold shares can sell call options to earn premium income. Example: A long-term investor may sell monthly covered calls, accepting the possibility that shares may be called away.

22.3 Speculation

Traders purchase calls when they expect prices to increase and puts when they expect prices to decrease. Example: A trader expecting NIFTY to rise following favourable economic announcements may purchase a call option.

Managing Volatility Risk

Options can be used to benefit from substantial market movements without predicting direction. Example: Before a major central-bank announcement, an investor may use a long straddle if a sufficiently large price movement is expected.

Corporate Risk Management

Companies use currency and commodity options to manage fluctuations in exchange rates and commodity prices. Example: An Indian company importing machinery from the United States must pay USD 500,000 after three months. If the rupee depreciates, the import cost increases. The company can purchase a USD call / INR put currency option to obtain protection against adverse exchange-rate movements, depending on the quotation convention and contract structure.

Employee Compensation

Companies provide stock options to employees as part of compensation packages. Employees receive an opportunity to purchase shares at a predetermined price, subject to vesting and other conditions. This can encourage retention and align employee interests with company performance.

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